# LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

## Quick Revision: Metals and Non-metals

### Physical properties

| Metals | Non-metals |
|---|---|
| Lustrous, hard, malleable, ductile, sonorous | Dull, brittle (solids), not sonorous |
| Good conductors of heat and electricity | Poor conductors |
| Mostly solids with high melting points | Solids, liquid (bromine) or gases |

**Exceptions:** mercury is a liquid metal. Sodium and potassium are soft and can be cut with a knife. Gallium and caesium melt on the palm. Iodine is lustrous. Diamond (carbon) is the hardest natural substance. Graphite conducts electricity.

### Chemical properties

- Metal + oxygen → **basic** oxide: $2Cu + O_2 \rightarrow 2CuO$. Non-metal oxides are acidic or neutral.
- **Amphoteric** oxides ($Al_2O_3$, ZnO) react with both acids and bases: $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$
- With water: Na and K react violently with cold water: $2Na + 2H_2O \rightarrow 2NaOH + H_2$ + heat. Ca reacts less violently. Mg reacts with hot water. Al, Fe and Zn react only with steam: $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$. Pb, Cu, Ag and Au do not react.
- With dilute acids: metals above hydrogen give $H_2$. Nitric acid is an oxidising agent, so usually no $H_2$ (except Mg and Mn with very dilute $HNO_3$).
- Displacement: a more reactive metal displaces a less reactive one: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

**Reactivity series:** K > Na > Ca > Mg > Al > Zn > Fe > Pb > [H] > Cu > Hg > Ag > Au

### Ionic compounds (e.g. NaCl, $MgCl_2$)

- Formed by transfer of electrons. $Na \rightarrow Na^+ + e^-$ and $Cl + e^- \rightarrow Cl^-$
- Hard solids with high melting and boiling points (strong forces between ions).
- Soluble in water, insoluble in kerosene and petrol.
- Conduct electricity when molten or in solution (ions can move), not when solid.

### Extraction of metals

| Reactivity | Method | Example |
|---|---|---|
| Low (Hg, Cu) | Heating alone | $2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$; $2HgO \rightarrow 2Hg + O_2$ |
| Middle (Zn, Fe, Pb) | Roasting (sulphide) or calcination (carbonate), then reduction with carbon | $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$; $ZnCO_3 \rightarrow ZnO + CO_2$; $ZnO + C \rightarrow Zn + CO$ |
| High (Na, Mg, Ca, Al) | Electrolysis of molten chloride/oxide | Na at cathode, $Cl_2$ at anode |

**Thermite reaction:** $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$ + heat (joins railway tracks). **Electrolytic refining:** impure metal is the anode, a thin strip of pure metal is the cathode, and impurities settle as anode mud.

### Corrosion & alloys

- Silver turns black (silver sulphide). Copper gets a green coat (basic copper carbonate). Iron rusts (needs both air and water).
- Prevention: painting, oiling, greasing, galvanising (zinc coat), chrome plating, anodising, alloying.

- Alloys: brass (Cu + Zn), bronze (Cu + Sn), solder (Pb + Sn, low melting point), stainless steel (Fe + Ni + Cr). 22 carat gold = 22 parts gold + 2 parts Cu or Ag.

## Quick check

1. Why is sodium kept under kerosene?

2. Name a metal that is extracted by electrolysis and one that is obtained by heating alone.

3. Give two conditions needed for iron to rust.